Model Paper - Database Management Systems

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Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)

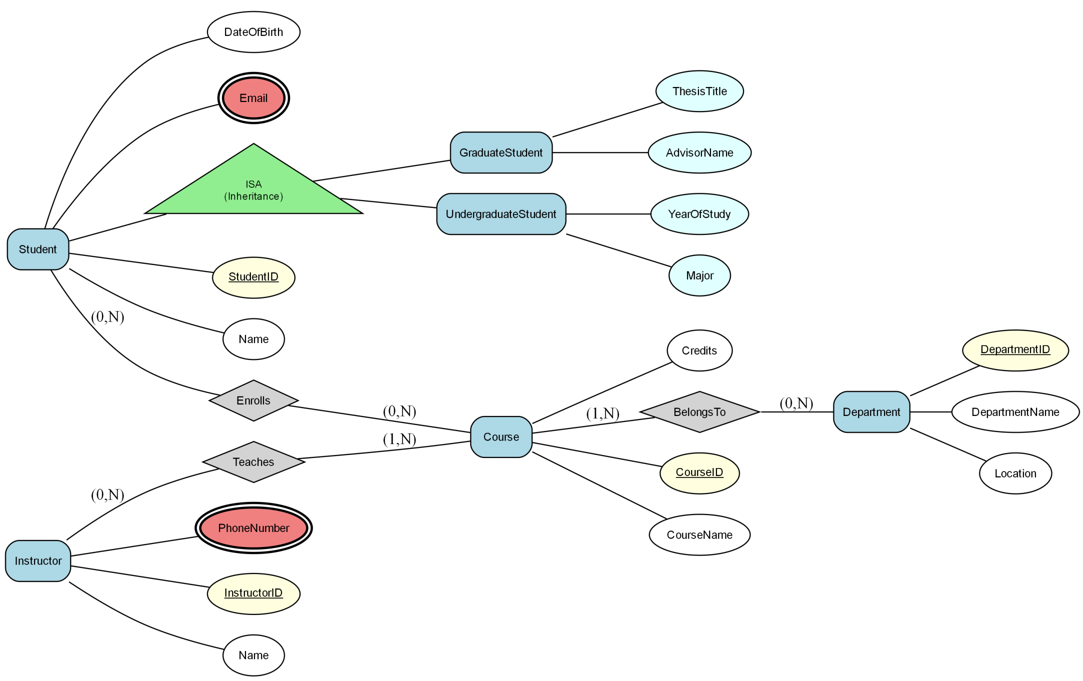
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)

- (0,1) indicates optional participation (zero or one)

- (0,N) or (0,*) indicates optional many participation (zero or many)

A university is managing its academic records for various programs. The entities involved are: 1) Student (StudentID, Name, DateOfBirth, Email), 2) Course (CourseID, CourseName, Credits), 3) Instructor (InstructorID, Name, PhoneNumber), and 4) Department (DepartmentID, DepartmentName, Location). Students can enroll in multiple courses and each course is taught by an instructor belonging to a specific department. The university has a specialization model where The Student entity has attributes such as StudentID, Name, Email. is further divided into Undergraduate and Graduate students. Graduate students have additional attributes: ThesisTitle and AdvisorName, while Undergraduate students carry attributes like YearOfStudy and Major. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional). A Instructor can be related to many Courses through the 'Teaches' relationship (Instructor participation is optional). Many Courses can be related to one Department through the 'BelongsTo' relationship (Department participation is optional)

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation R(A, B, C, D, E) with the following set of functional dependencies F over R: F={{AB->C, C->D, D->E, B->A}}. Analyze the schema for normalization.

a) Find all the keys in relation R using the attribute closure method. (4 marks)

b) Is R in 3NF? Give reasons for your conclusion. (3 marks)

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is designing a relational database to manage student enrollment, course offerings, and grades. The database schema includes tables such as Students, Courses, and Enrollments. Each table is defined with specific attributes: Students have attributes like StudentID, Name, and DateOfBirth; Courses have CourseID, Title, and Credits; and Enrollments contain EnrollmentID, StudentID, CourseID, and Grade. The university plans to ensure data integrity and efficient data retrieval through appropriate SQL DDL and DML operations.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: ```java String sql = "SELECT * FROM employees WHERE department = ?"; PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs = pstmt.executeQuery(); ``` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the role of foreign keys in the Enrollments table and their impact on the database structure. (2 marks)

d) Briefly explain the significance of the SQL command used to delete a course from the Courses table while ensuring that no orphaned records exist. (3 marks)

e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing

the Enrollments database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for an E-commerce platform: Product (productId: int, productName: varchar(100), price: decimal(10, 2), stockQuantity: int) Customer (customerId: int, customerName: varchar(100), email: varchar(50), phoneNumber: varchar(15)) Order (orderId: int, orderDate: date, customerId: int, totalAmount: decimal(10, 2)) OrderItem (orderItemId: int, orderId: int, productId: int, quantity: int, itemTotal: decimal(10, 2)) The 'Product' table stores information about products available for sale, including unique product ID, product name, price, and stock quantity. The 'Customer' table holds customer details, which consists of a unique customer ID, name, email, and phone number. The 'Order' table keeps track of customer orders with order information, including unique order ID, order date, linked customer ID, and the total order amount. The 'OrderItem' table contains details of each product in an order, including order item ID, linked order ID, product ID, quantity ordered, and item total price. The 'Order' table manages order records, tracking customer orders with order ID, customer ID, order date, and total amount.

a) Write SQL Queries to perform the following: i. Find the product name and total amount for orders placed on '2023-09-15'.

i. Find the product name and total amount for orders placed on '2023-09-15'. (4 marks)

ii. Find the customer who has the highest total totalamount compared to all other customers. Find the customerId and customerName. (6 marks)

iii. Find the productName and customerName from Product, Customer, and Order where Order.orderDate is not null. (7 marks)

b) b) Create a function to calculate the total amount for a given customer. This function should account for customer with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'totalAmount' column in the 'Order' table whenever a new order record is added, or an existing order record is updated. The 'totalAmount' column should contain the total amount for each customer. Ensure the trigger adjusts the 'totalAmount' column accurately to reflect any partial payments made by customer. The customer pays only 50% of the total amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)